

Bitwise linear trail through **LBlock** found by SAT

CiVerLy

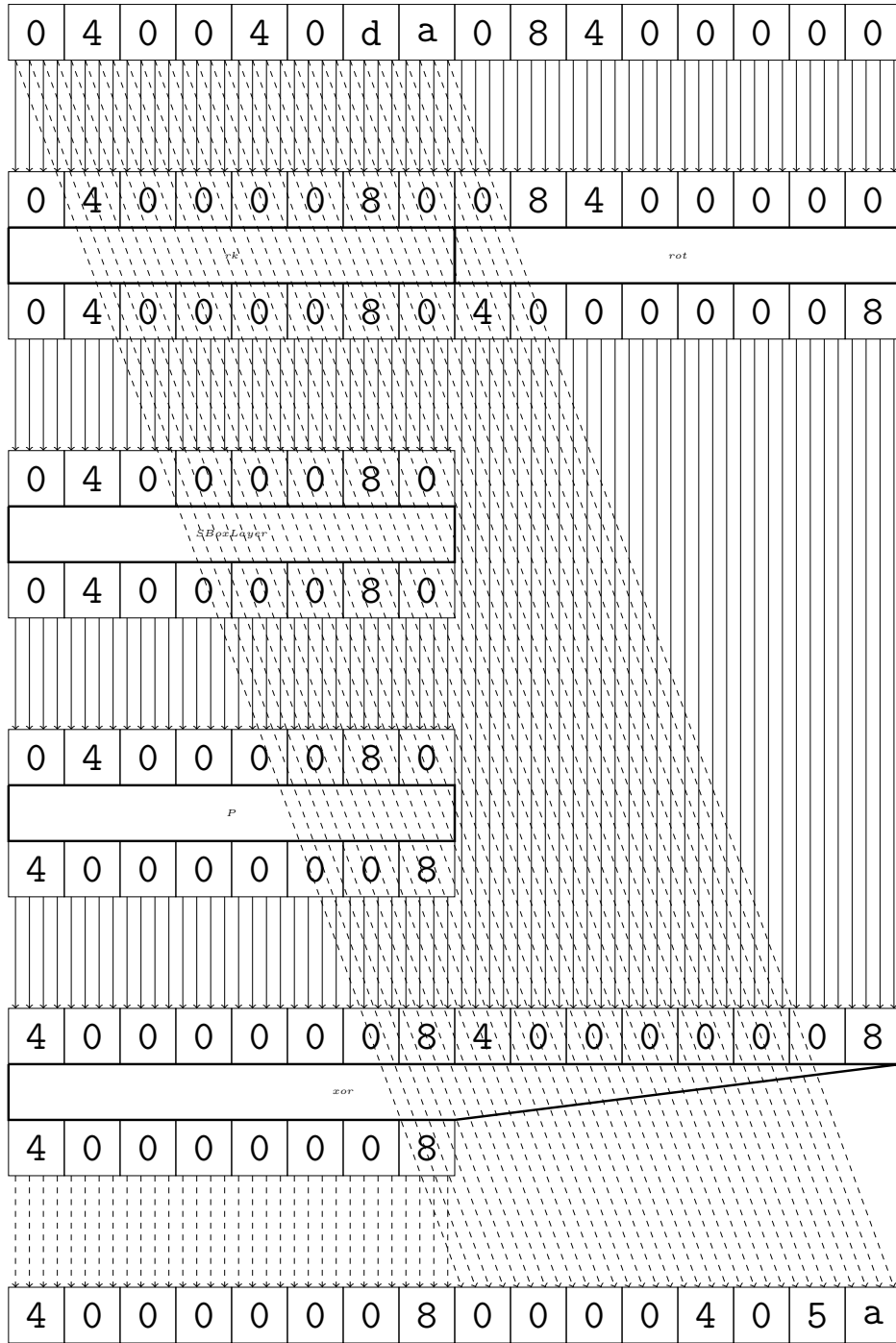
July 20, 2026

1 LBlock

Maximal linear correlation: 2^{-36}

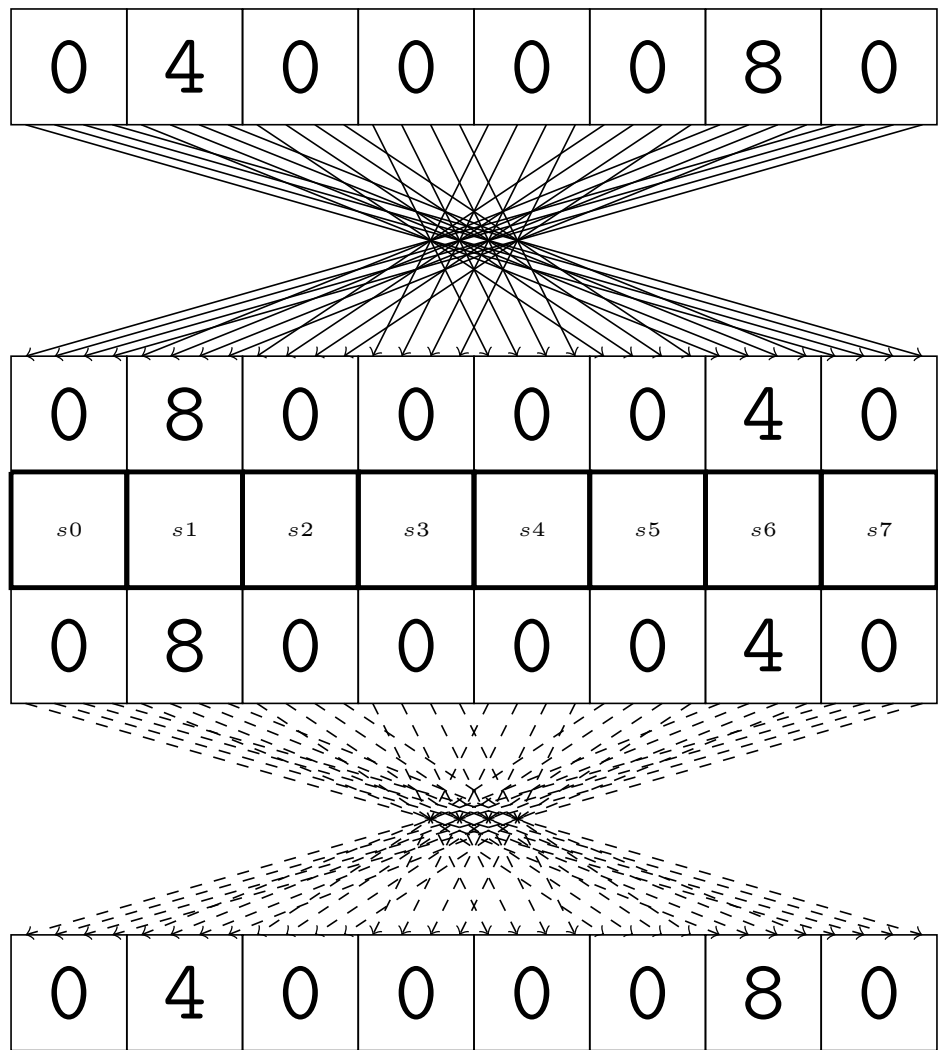
0	4	0	0	4	0	d	a	0	8	4	0	0	0	0
0	4	0	0	4	0	d	a	0	8	4	0	0	0	0
4	0	0	0	0	0	8	0	0	0	0	4	0	5	a
4	0	0	0	0	0	8	0	0	0	0	4	0	5	a
0	0	4	0	5	a	0	0	0	0	0	0	e	0	0
0	0	4	0	5	a	0	0	0	0	0	0	e	0	0
0	0	0	e	0	0	0	0	0	0	0	5	a	0	0
0	0	0	e	0	0	0	0	0	0	0	5	a	0	0
0	0	5	a	0	0	0	c	0	4	e	0	0	0	0
0	0	5	a	0	0	0	c	0	4	e	0	0	0	0
4	e	0	0	0	c	0	0	4	5	0	4	0	0	0
4	e	0	0	0	c	0	0	4	5	0	4	0	0	0
5	0	4	0	0	0	4	0	0	0	0	0	0	0	0
5	0	4	0	0	0	4	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	5	0	4	0	0	0	0	4
0	0	0	0	0	0	0	5	0	4	0	0	0	0	4
4	0	0	0	0	4	5	0	0	4	0	0	d	0	c
4	0	0	0	0	4	5	0	0	4	0	0	d	0	c
0	0	d	0	0	c	0	4	0	0	0	0	4	9	e
0	0	d	0	0	c	0	4	0	0	0	0	4	9	e
0	0	0	4	9	e	0	0	0	0	0	0	0	0	0
0	0	0	4	9	e	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	4	9	e	0
0	0	0	0	0	0	0	0	0	0	0	4	9	e	0
0	4	9	e	0	0	0	0	8	0	4	8	0	0	0
0	4	9	e	0	0	0	0	8	0	4	8	0	0	0
4	8	0	0	0	8	0	0	8	9	0	4	0	0	0
4	8	0	0	0	8	0	0	8	9	0	4	0	0	0
9	0	4	0	0	0	8	0	0	0	0	0	0	0	0
9	0	4	0	0	0	8	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	9	0	4	0	0	0	0	8
0	0	0	0	0	0	0	9	0	4	0	0	0	0	8
4	0	0	0	8	9	0	0	4	0	0	7	0	0	8
4	0	0	0	8	9	0	0	4	0	0	7	0	0	8
0	4	0	0	7	0	0	8	4	0	0	0	8	9	0
0	4	0	0	7	0	0	8	4	0	0	0	8	9	0

2 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

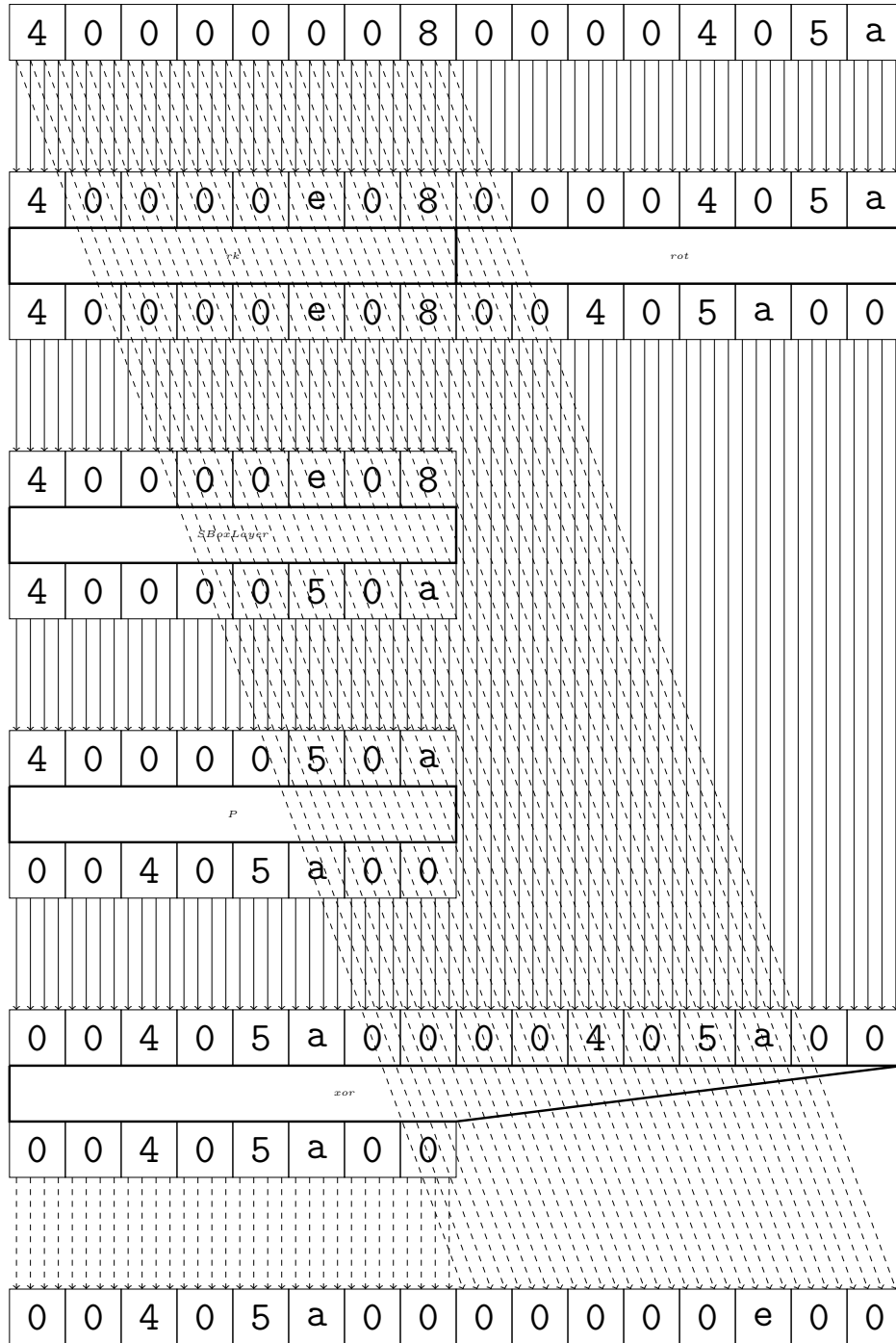
3 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



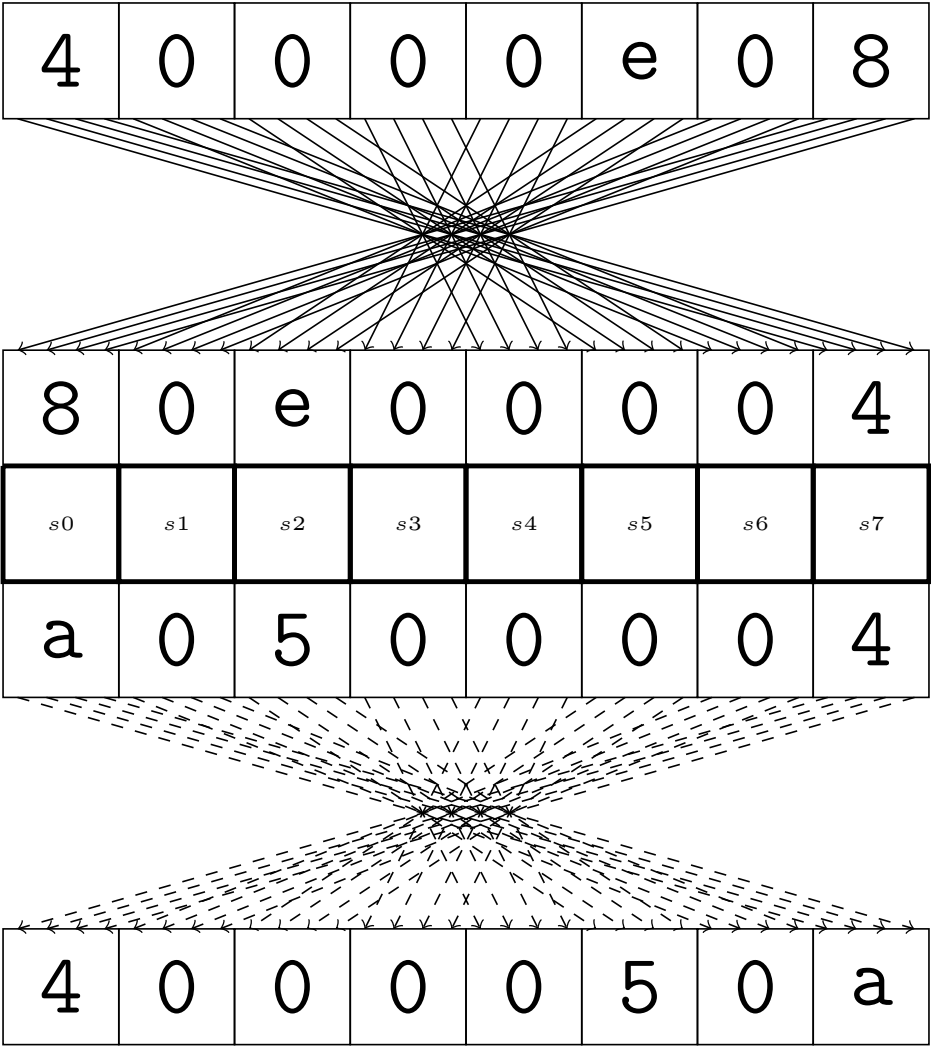
4 LBlock_round

Maximal linear correlation: $2^{-3.0}$



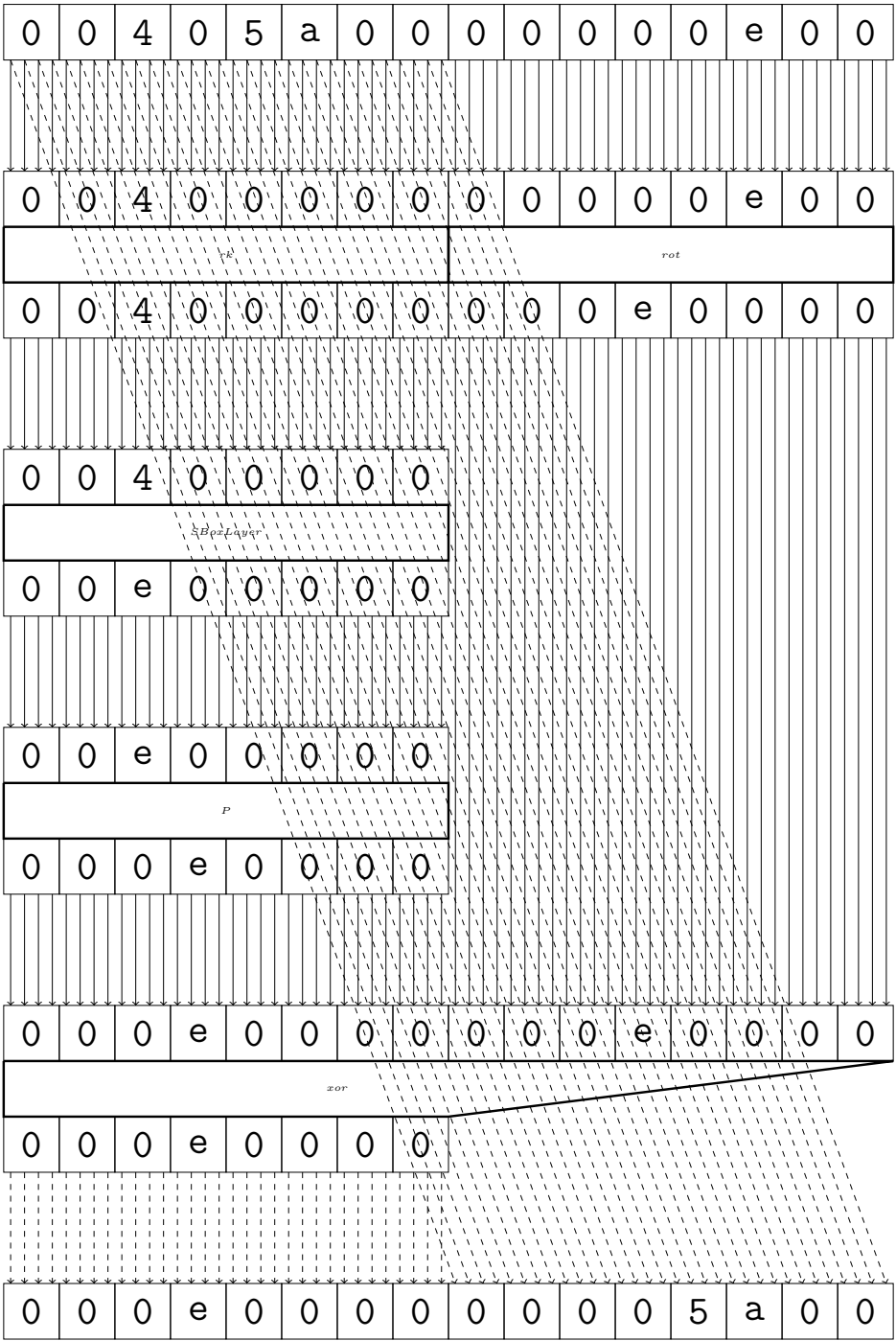
5 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



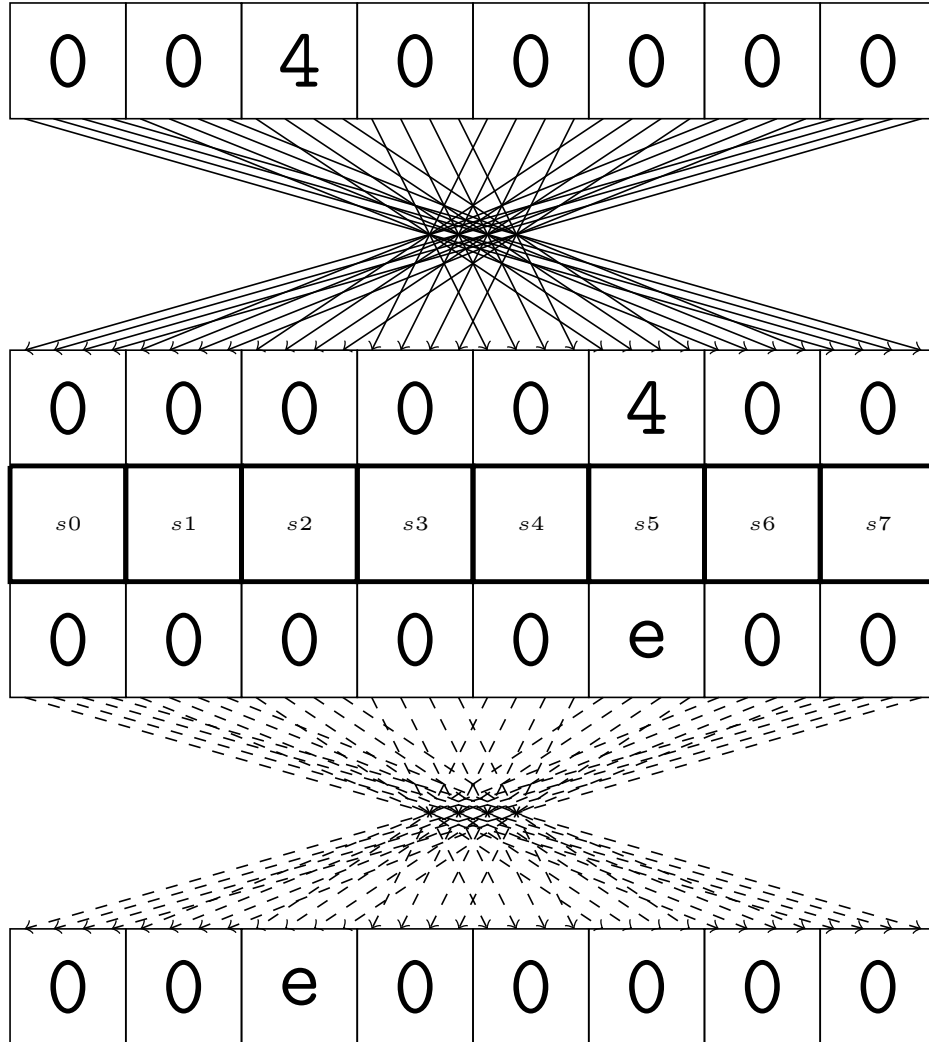
6 LBlock_round

Maximal linear correlation: $2^{-1.0}$



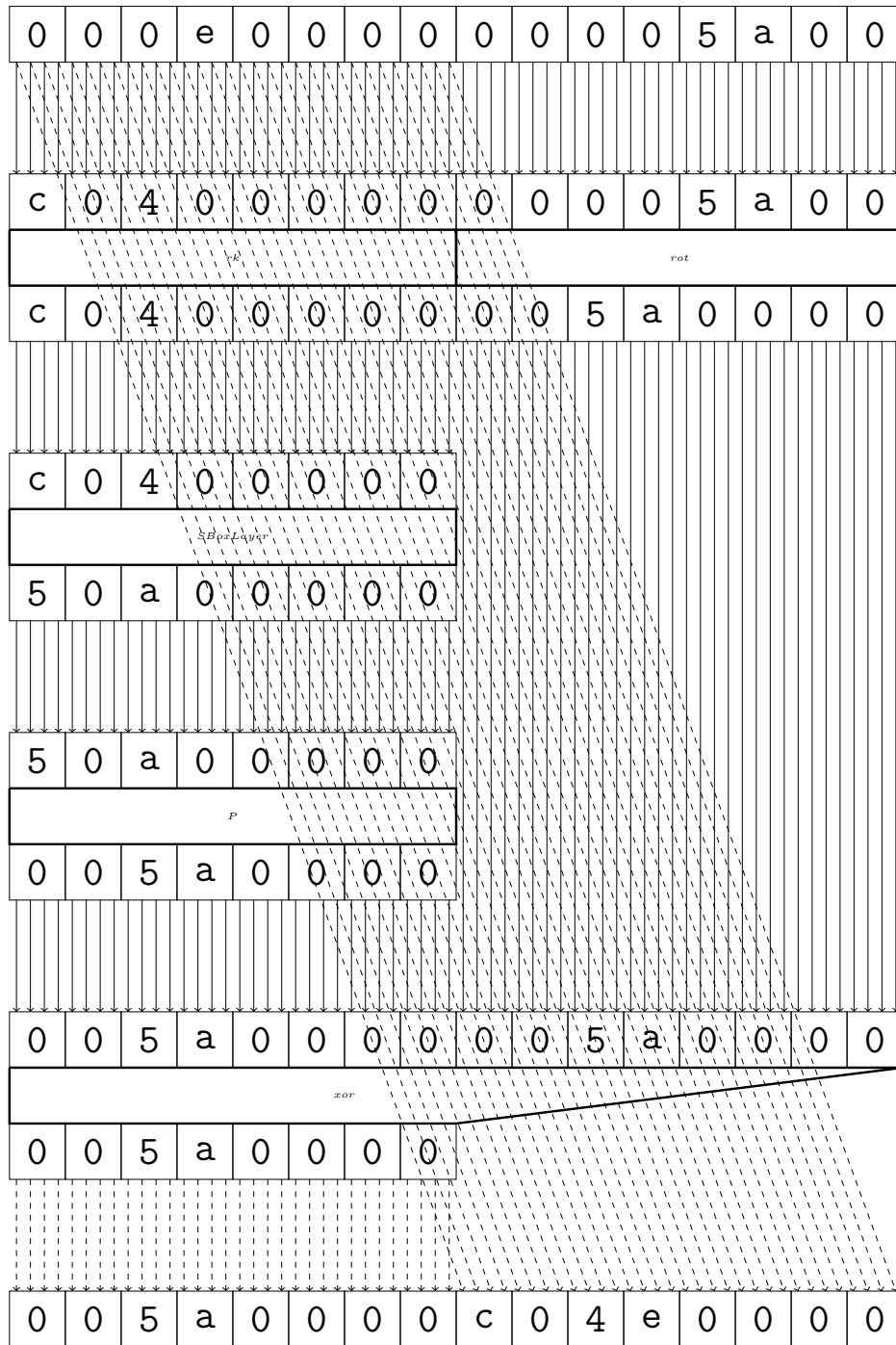
7 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



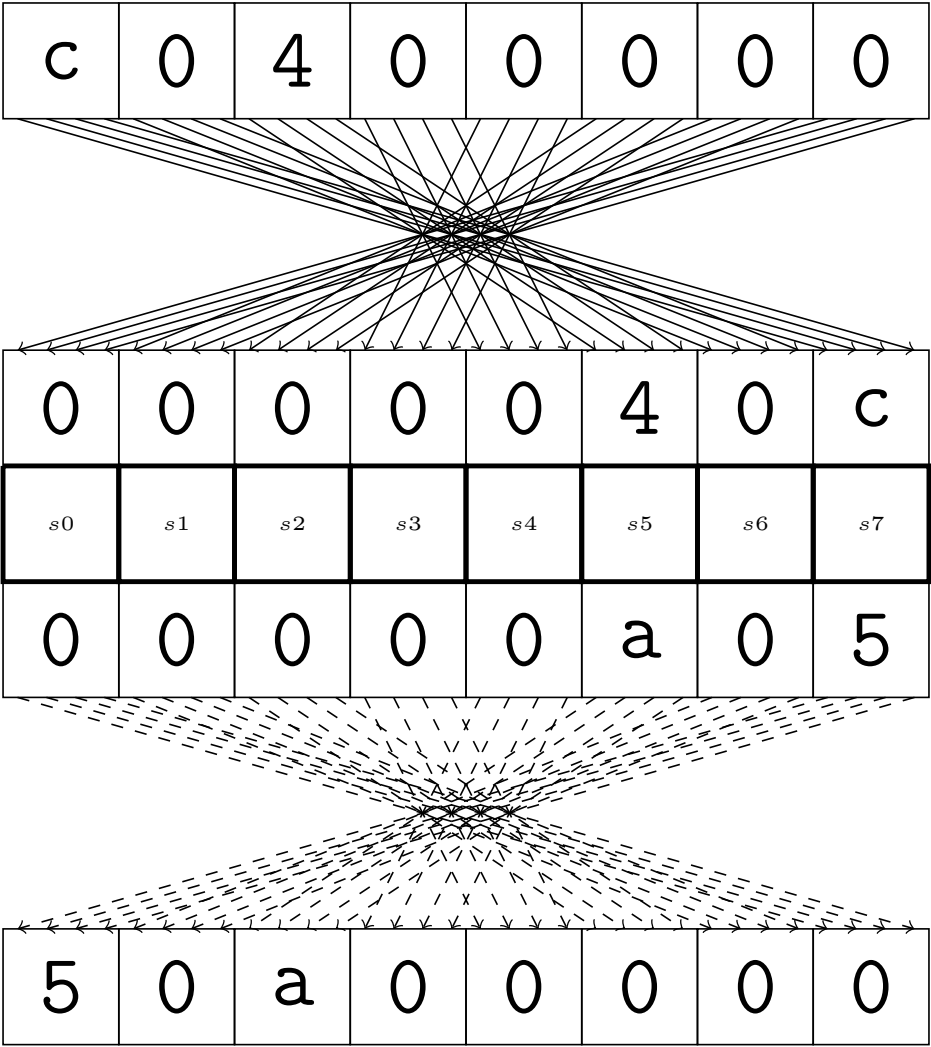
8 LBlock_round

Maximal linear correlation: $2^{-2.0}$

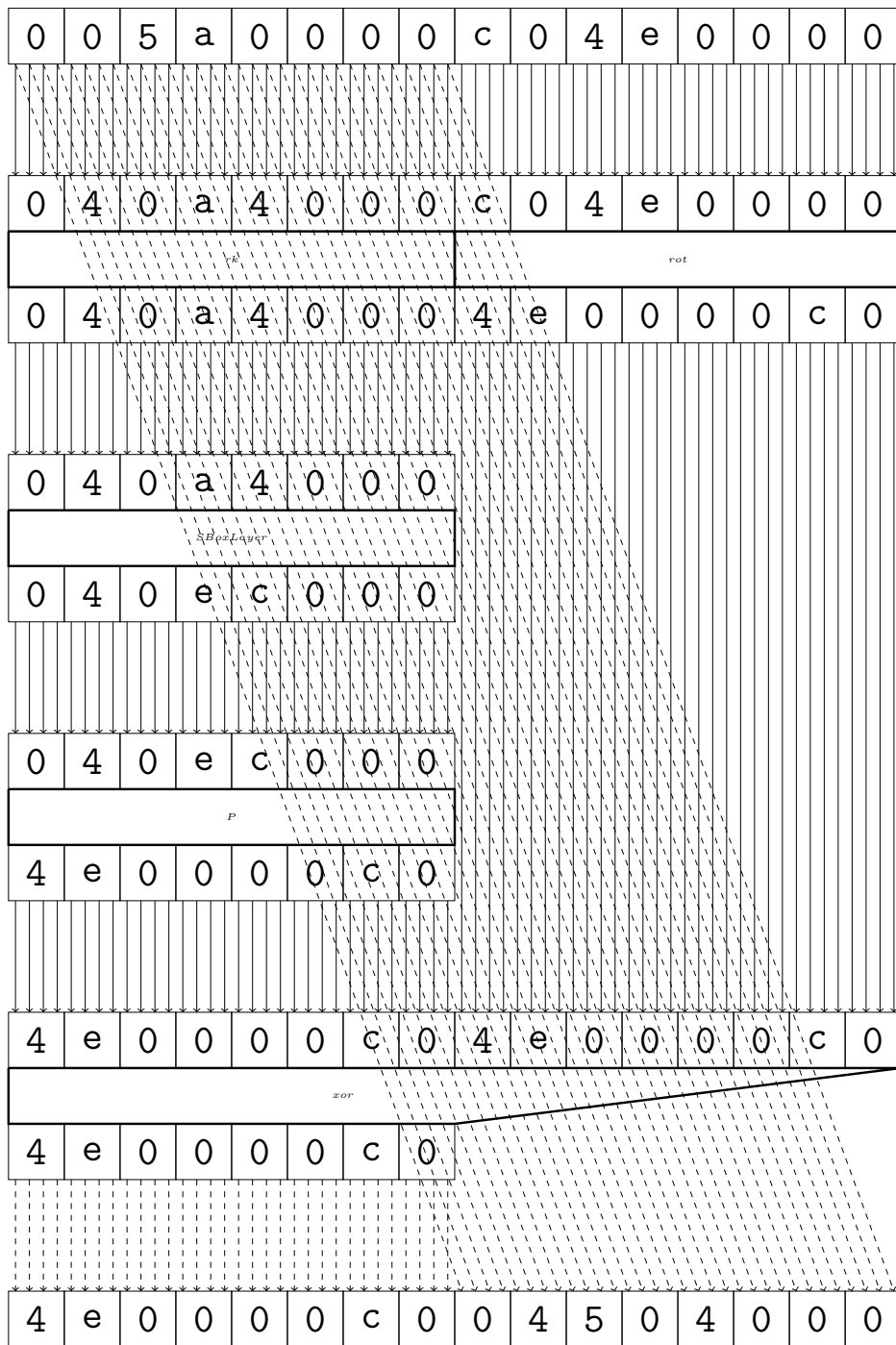


9 SBoxLayer

Maximal linear correlation: $2^{-2.0}$

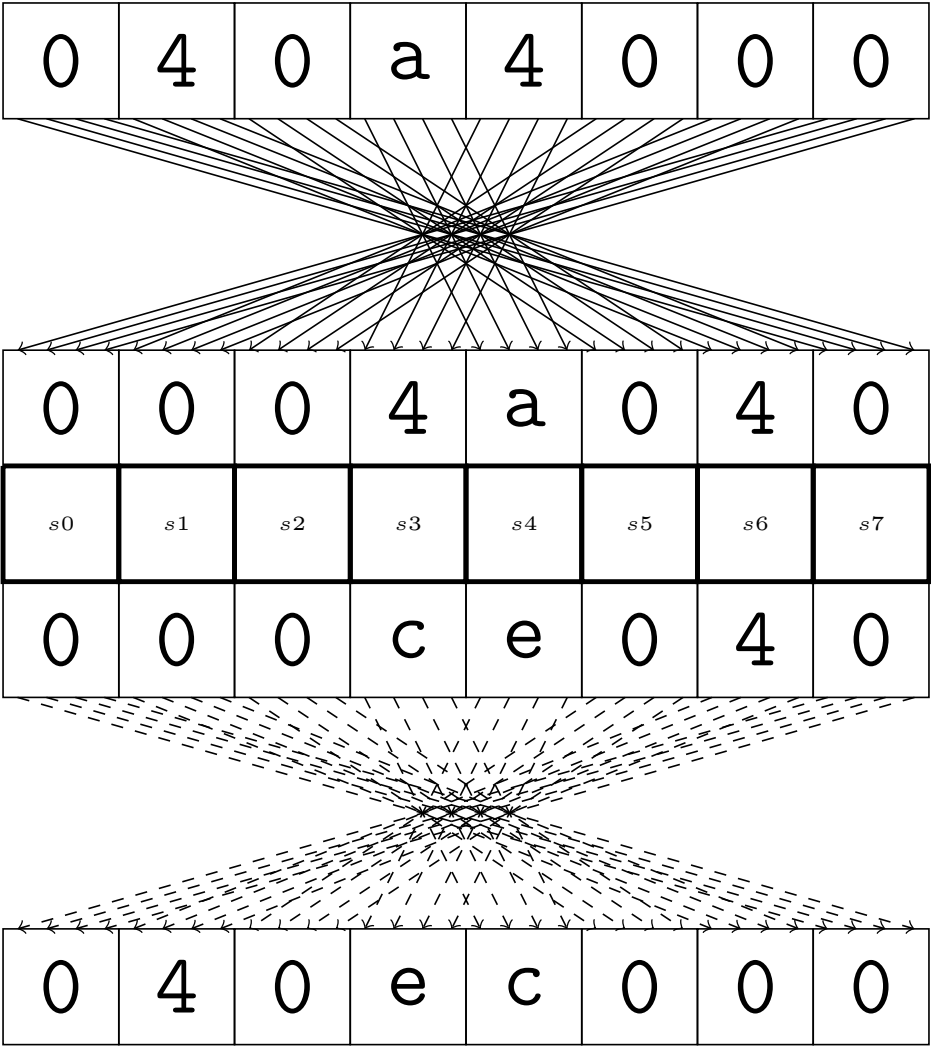


10 LBlock_round

Maximal linear correlation: $2^{-3.0}$ 

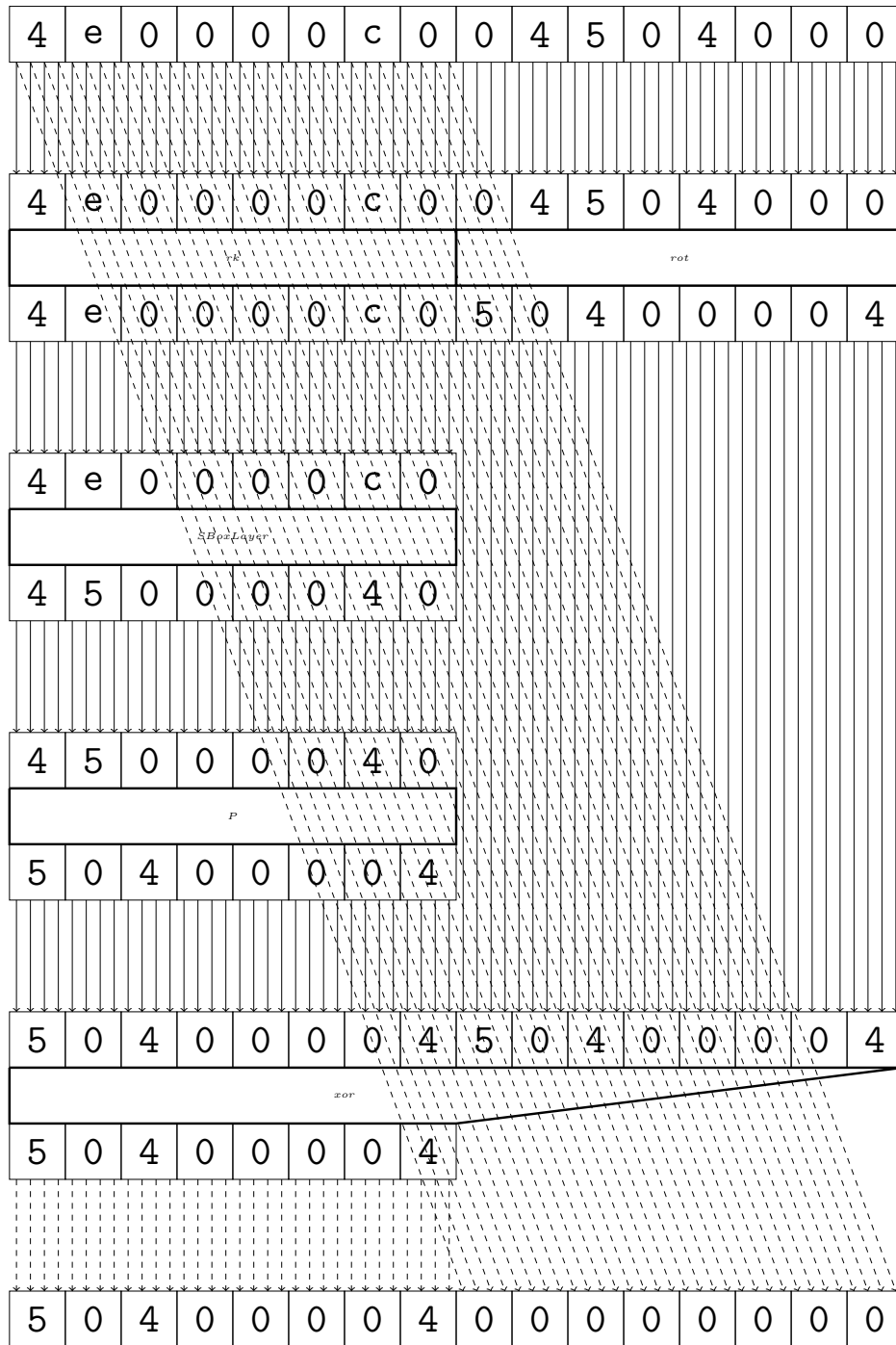
11 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



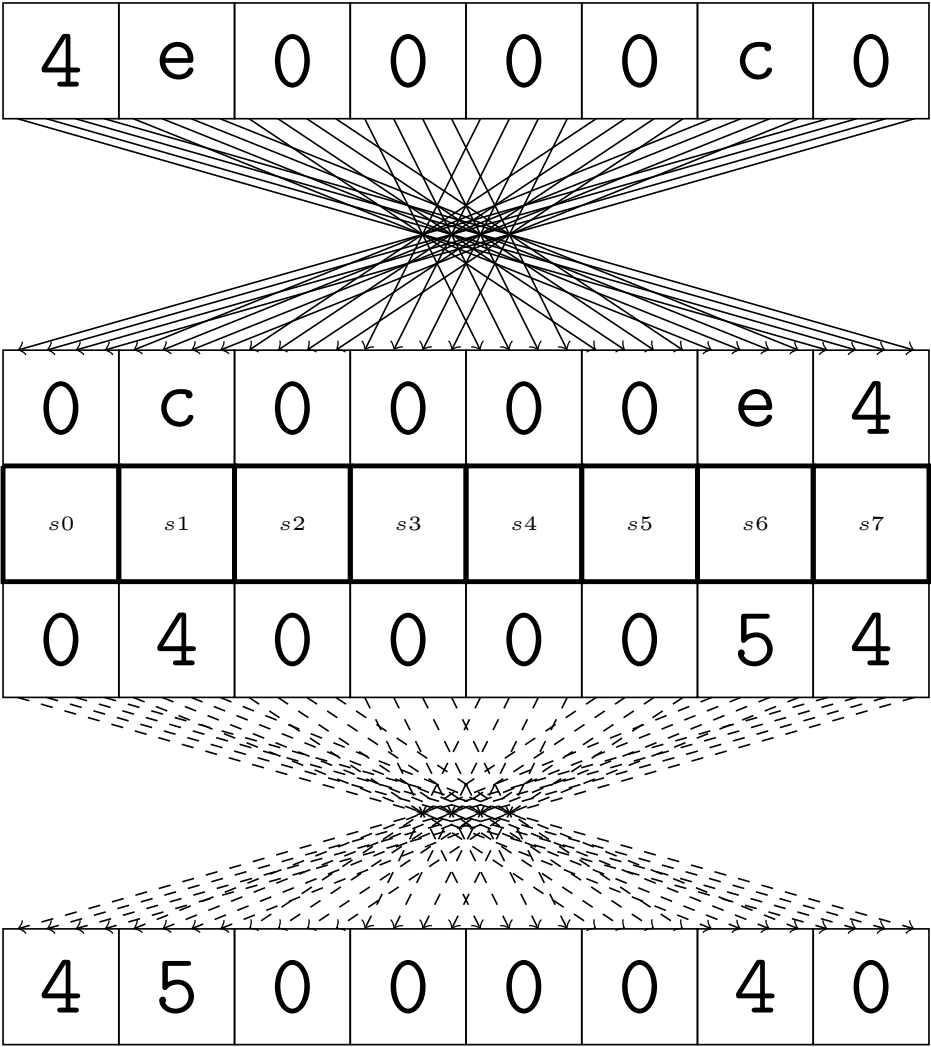
12 LBlock_round

Maximal linear correlation: $2^{-4.0}$

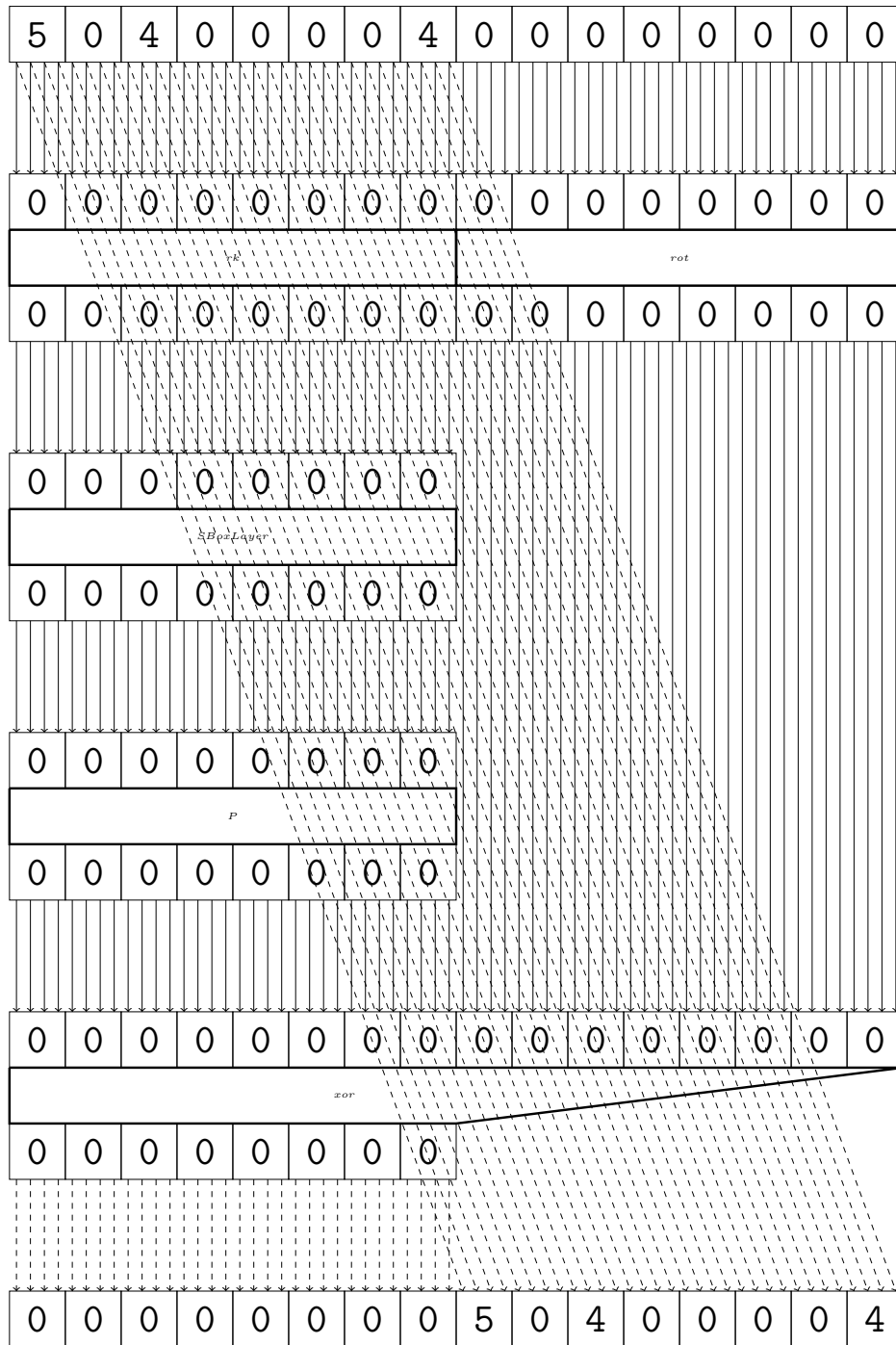


13 SBoxLayer

Maximal linear correlation: $2^{-4.0}$

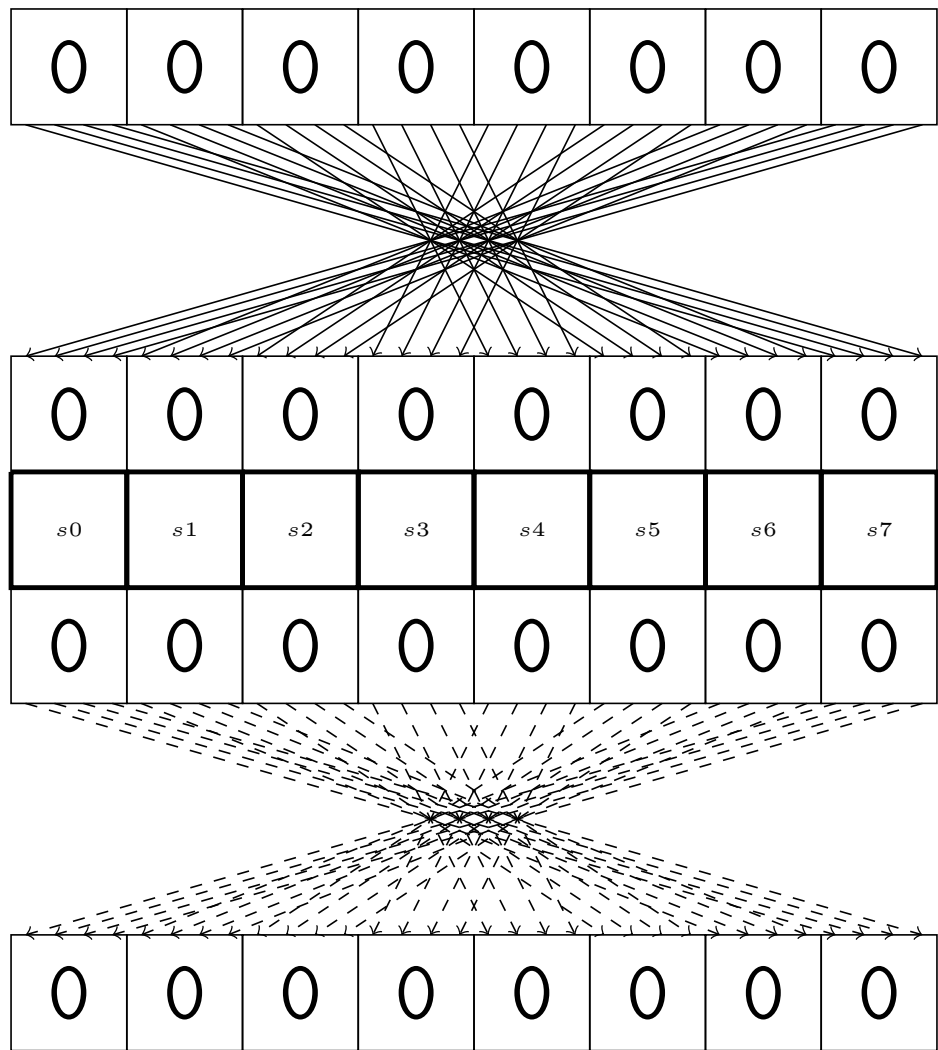


14 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

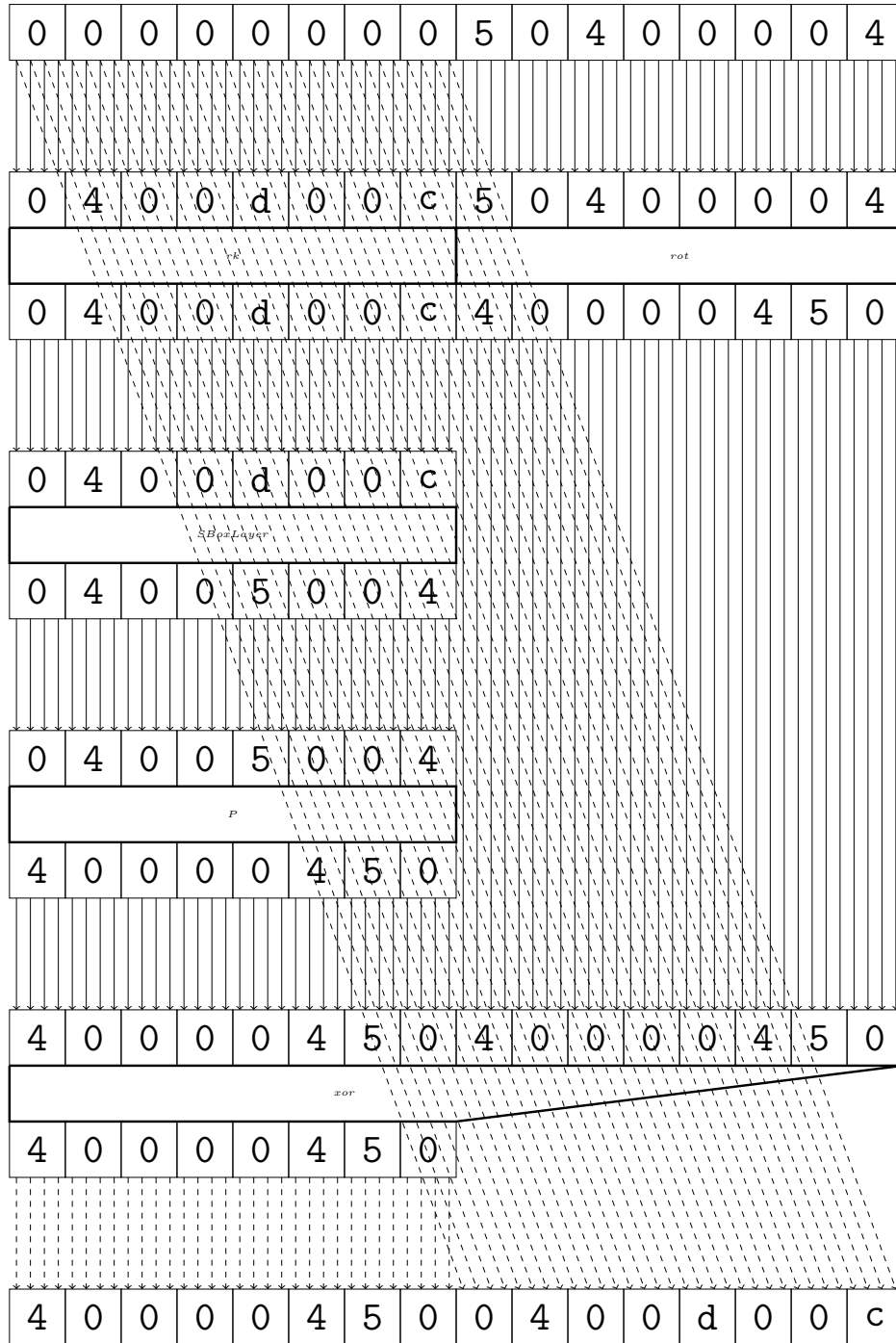
15 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



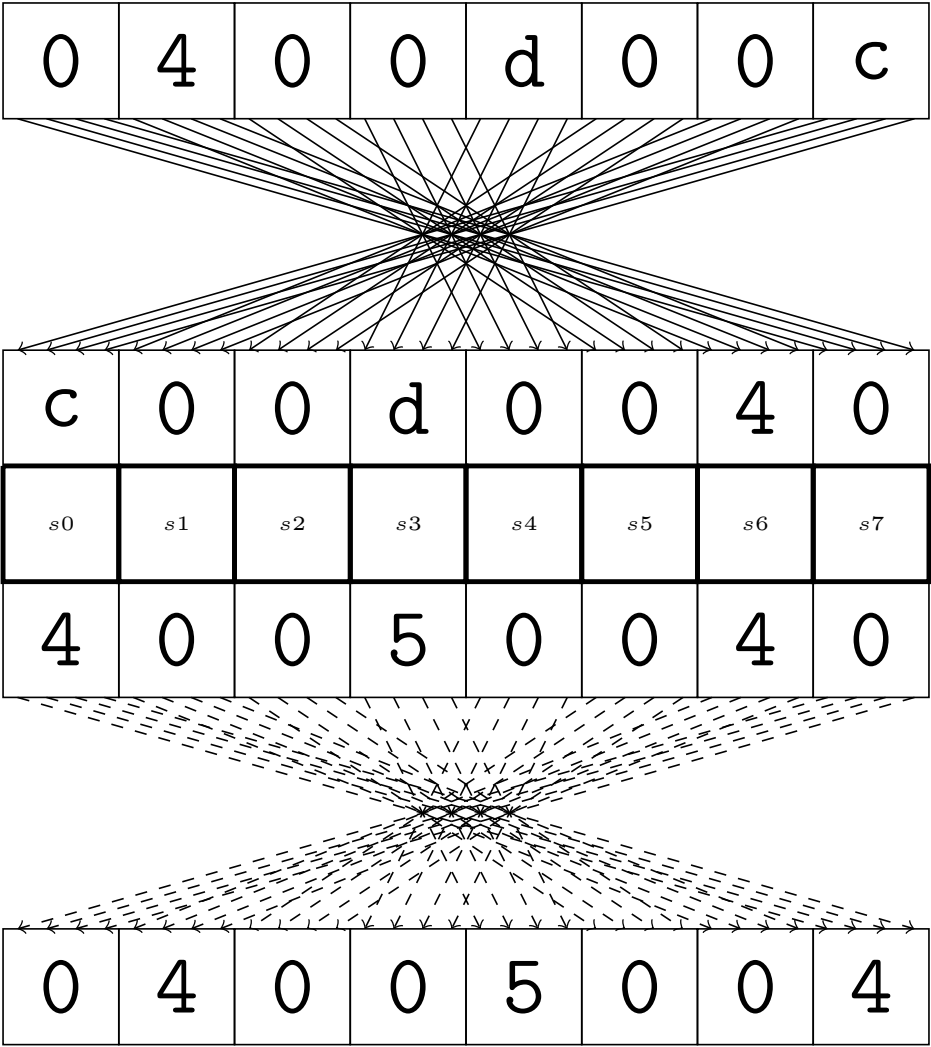
16 LBlock_round

Maximal linear correlation: $2^{-3.0}$

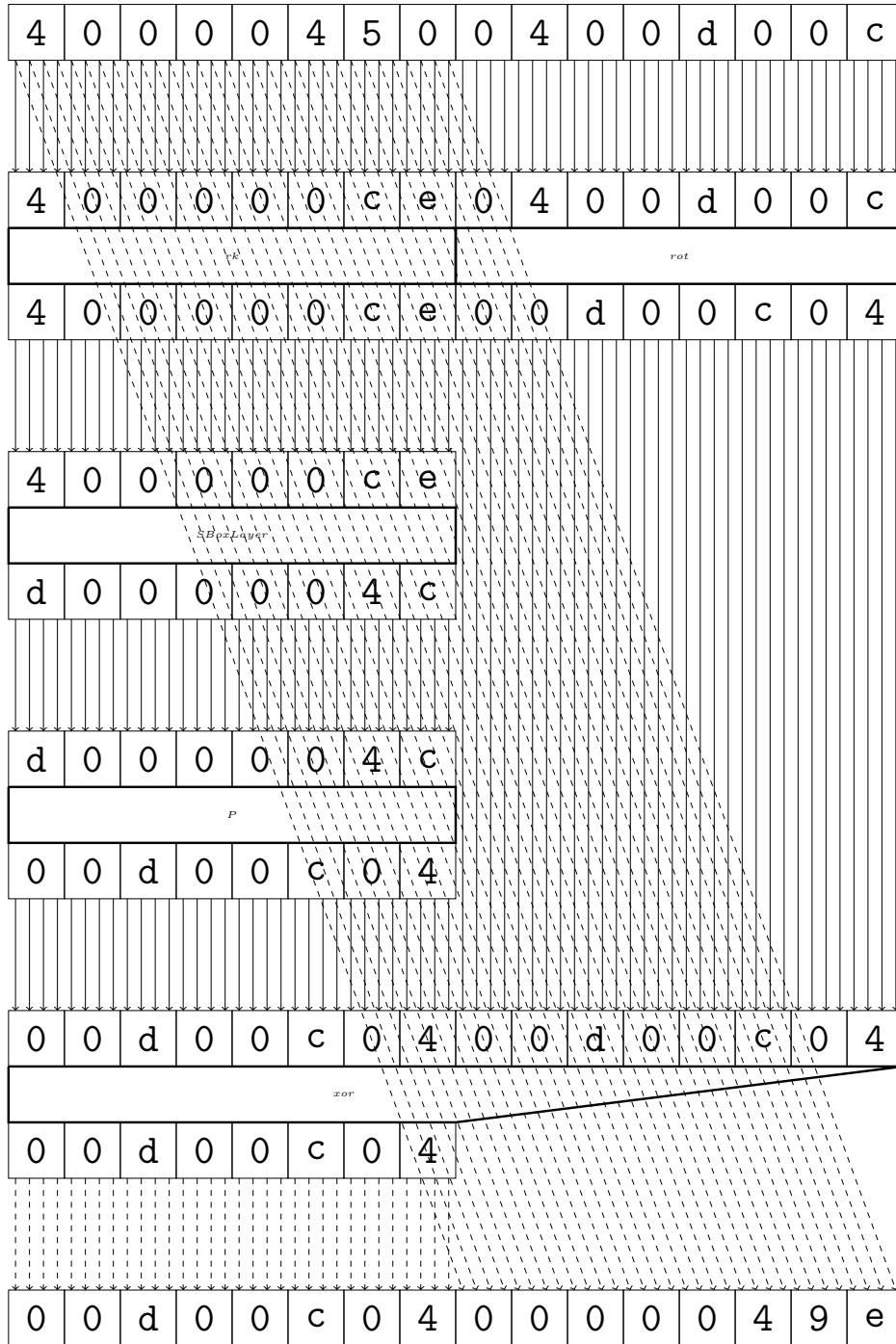


17 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

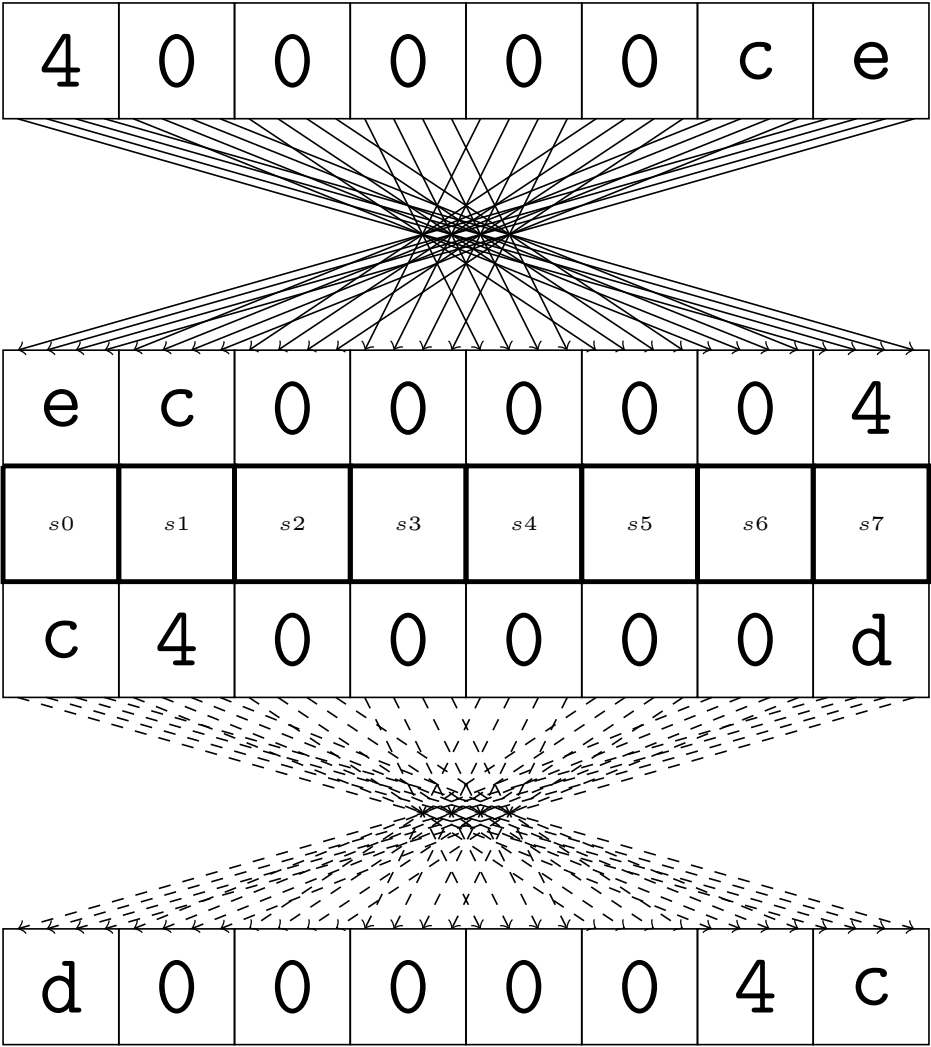


18 LBlock_round

Maximal linear correlation: $2^{-3.0}$ 

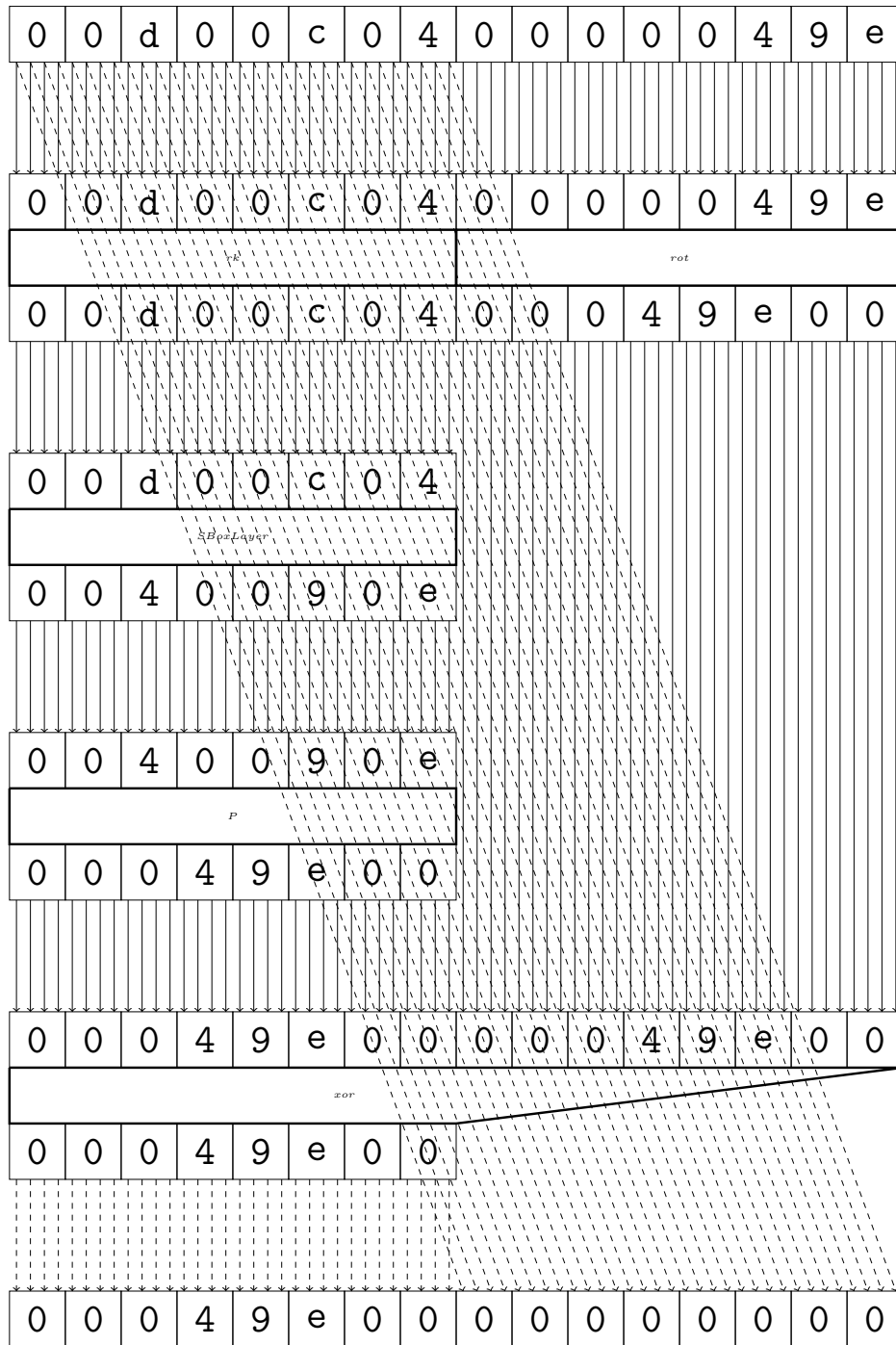
19 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



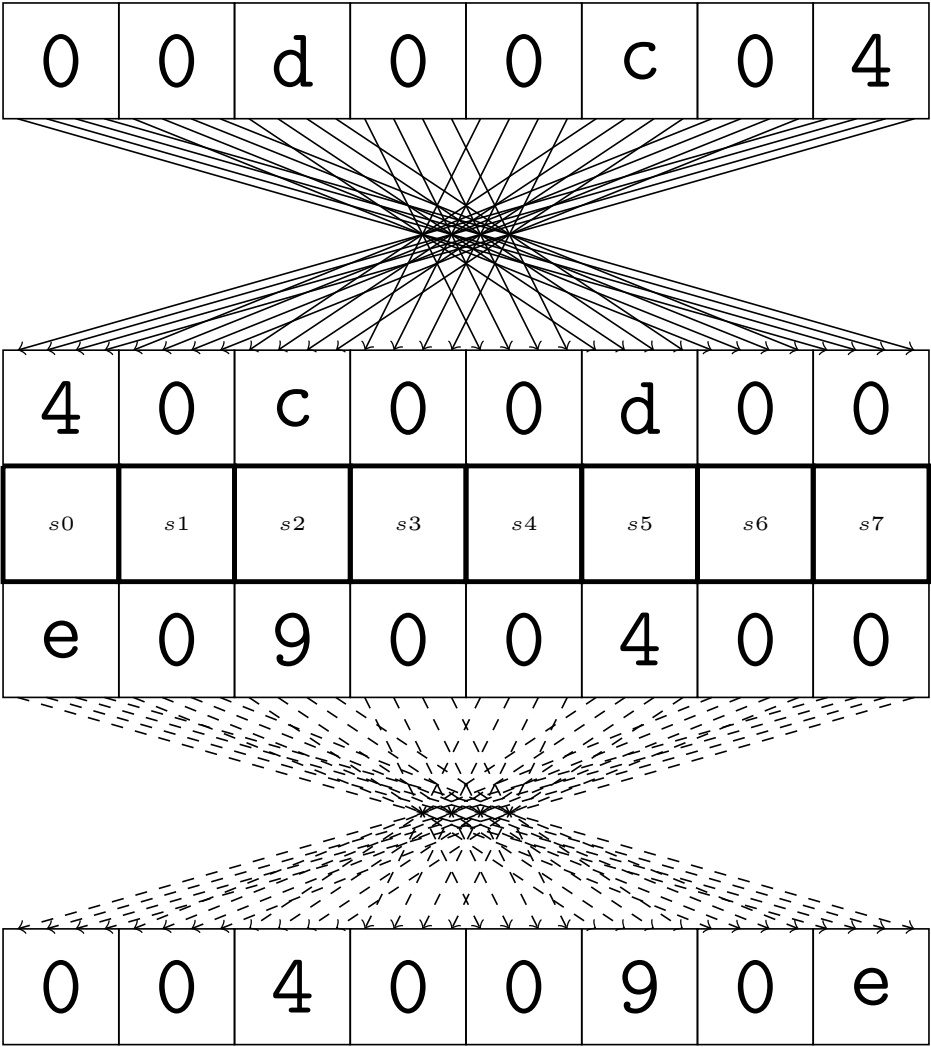
20 LBlock_round

Maximal linear correlation: $2^{-3.0}$

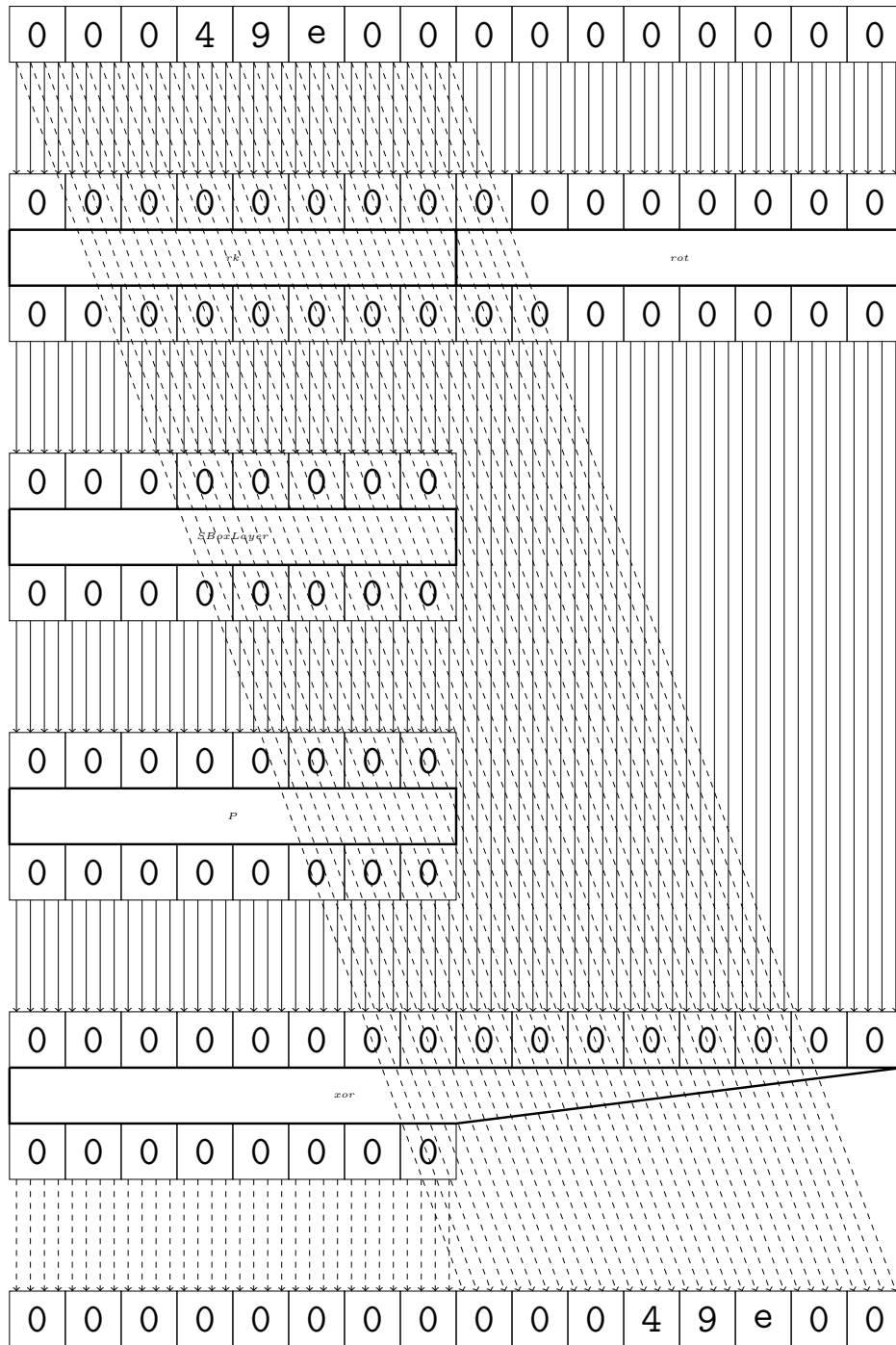


21 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

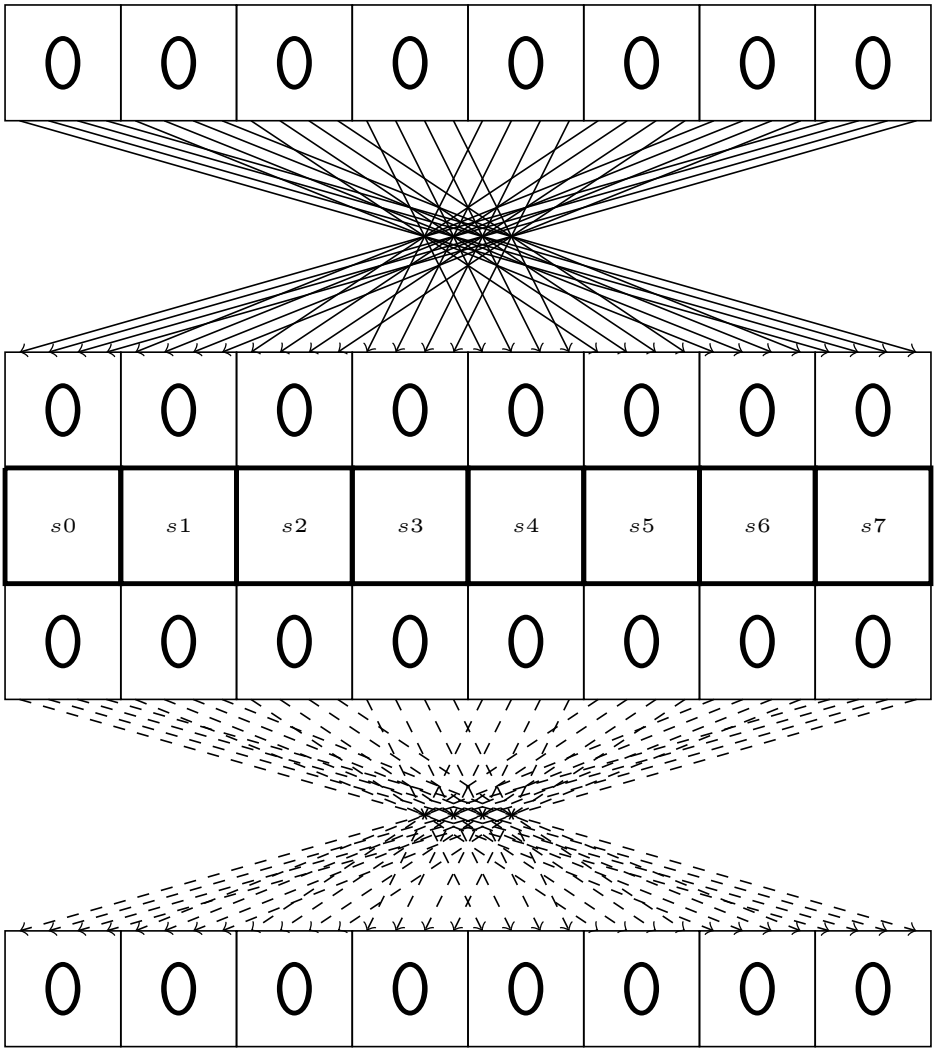


22 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

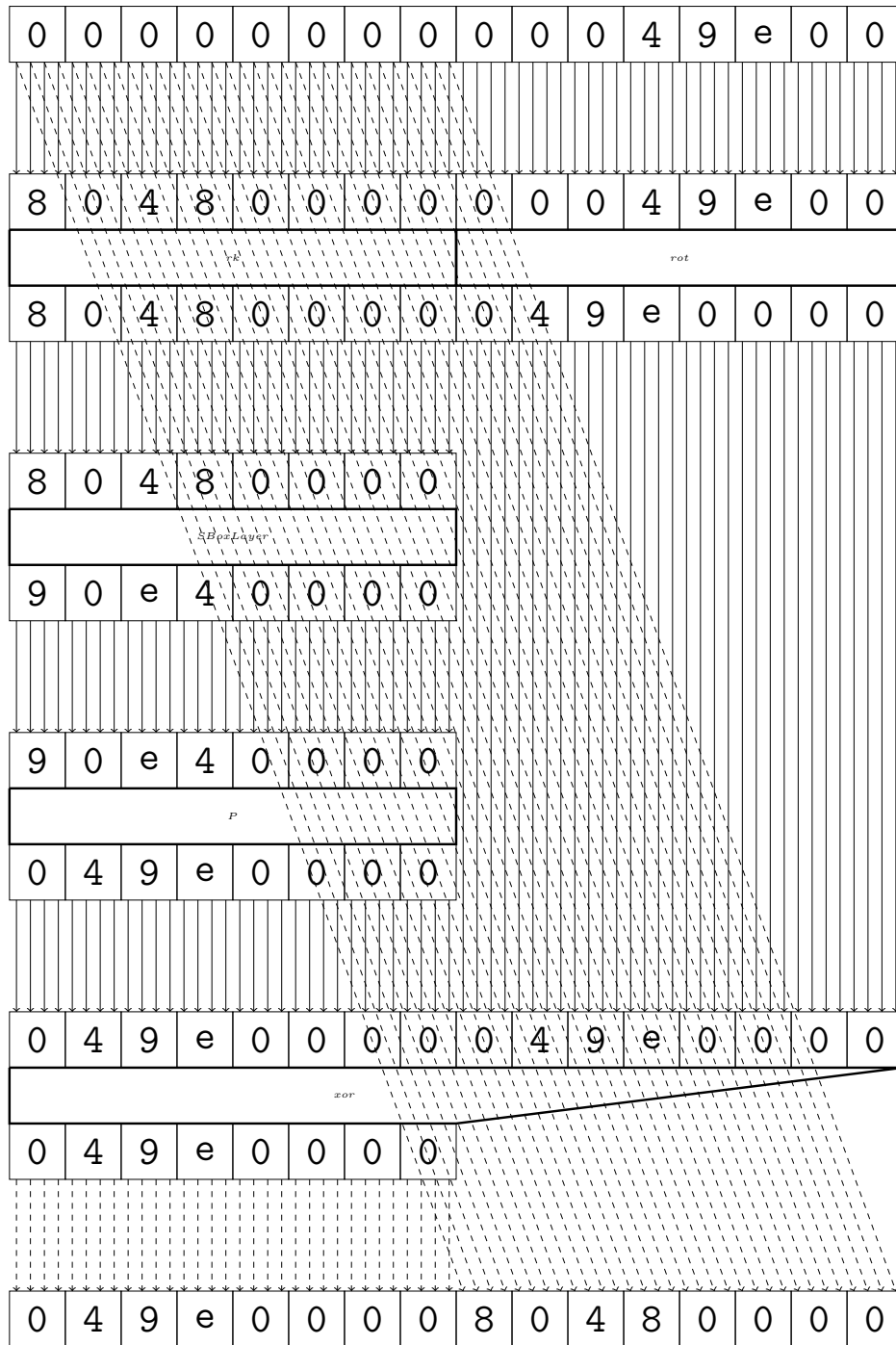
23 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



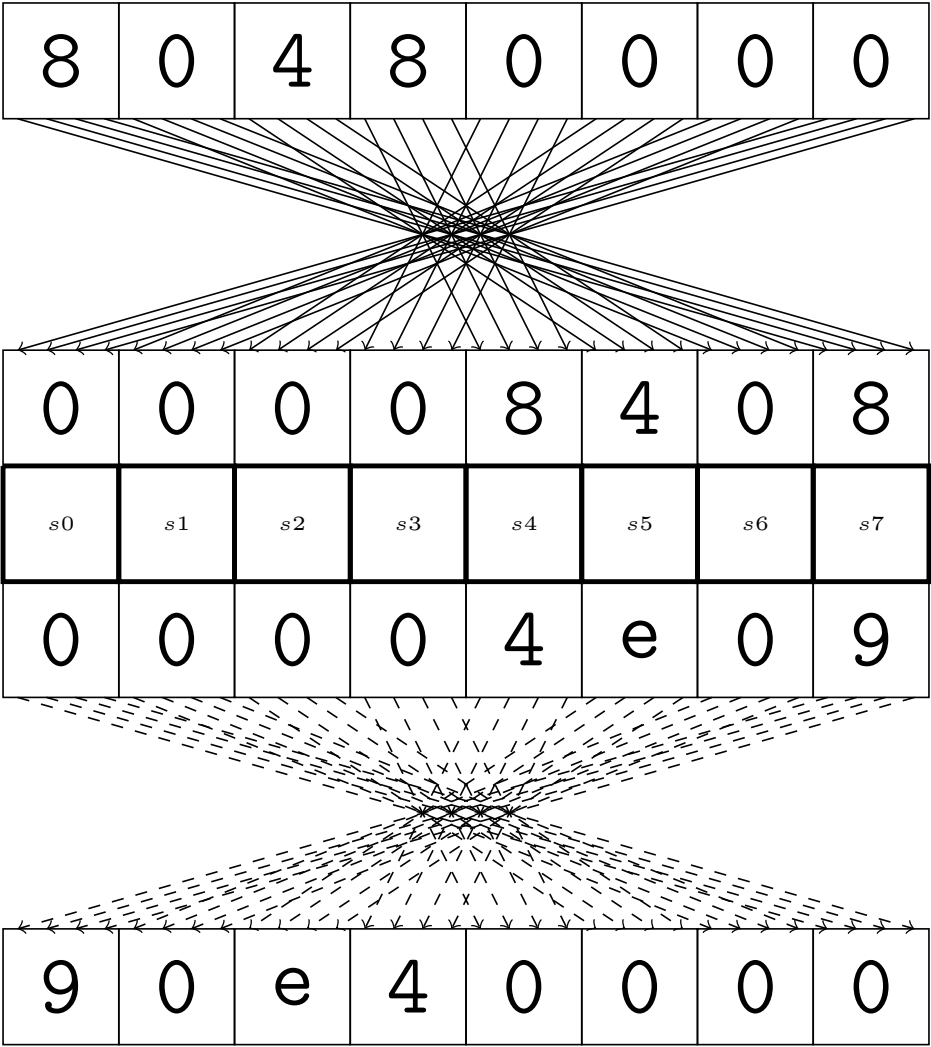
24 LBlock_round

Maximal linear correlation: $2^{-3.0}$



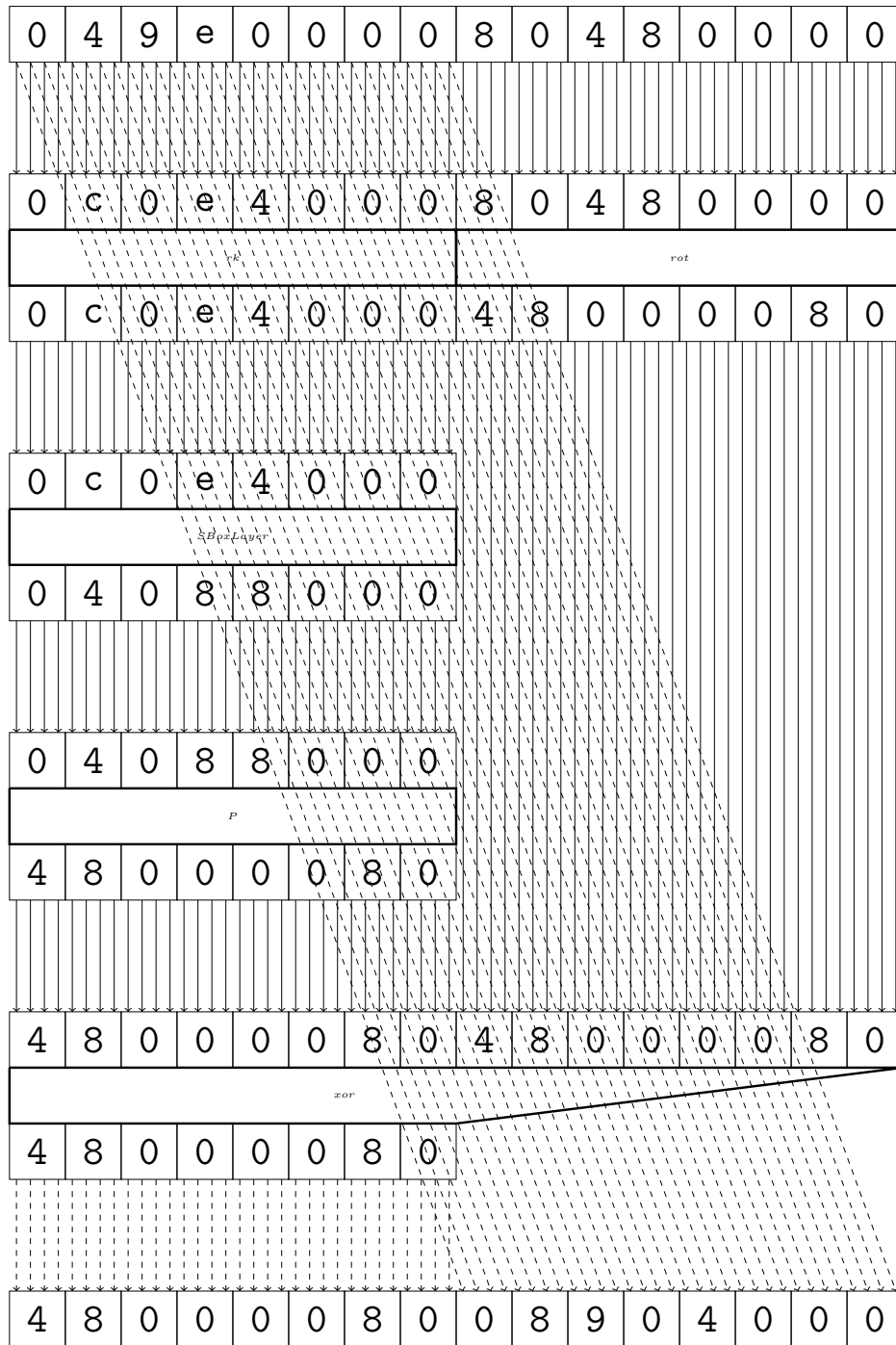
25 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



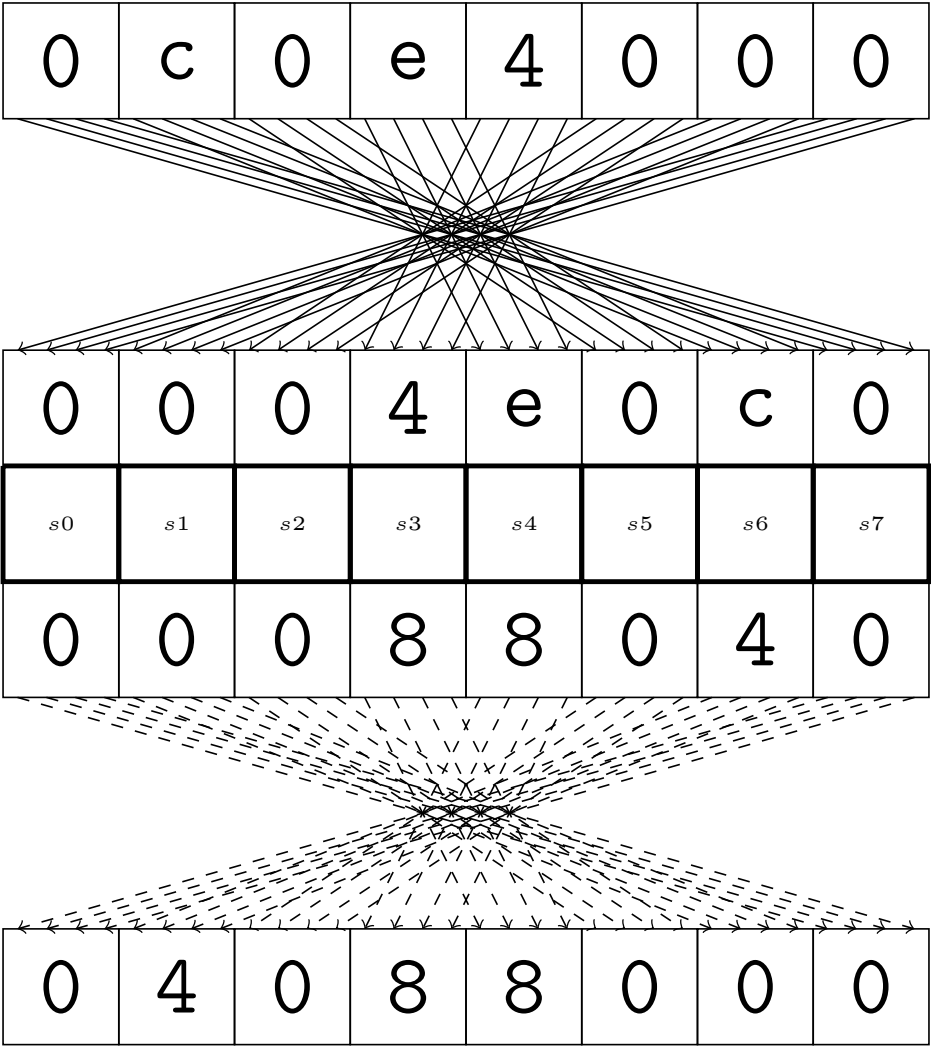
26 LBlock_round

Maximal linear correlation: $2^{-3.0}$



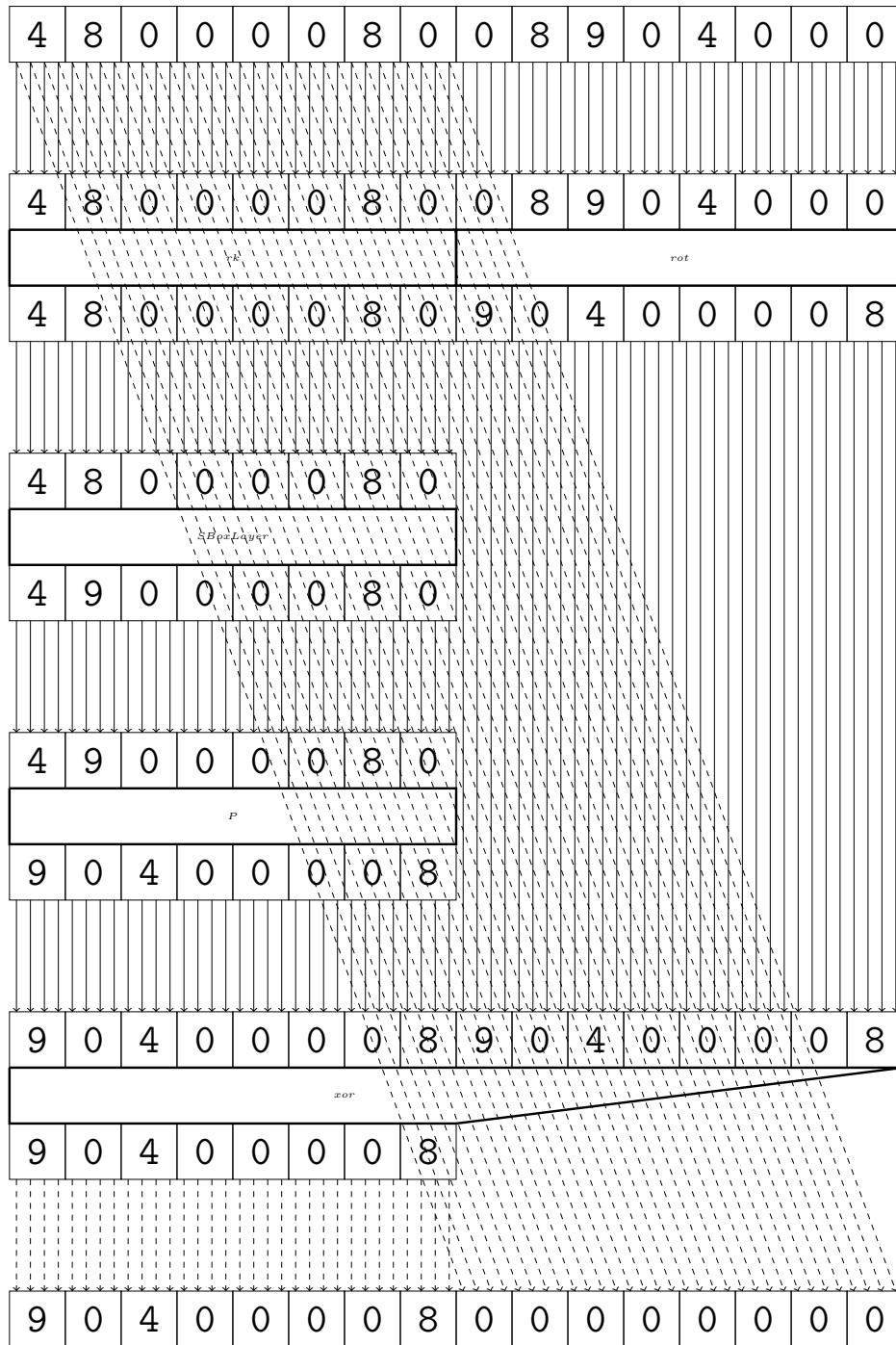
27 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



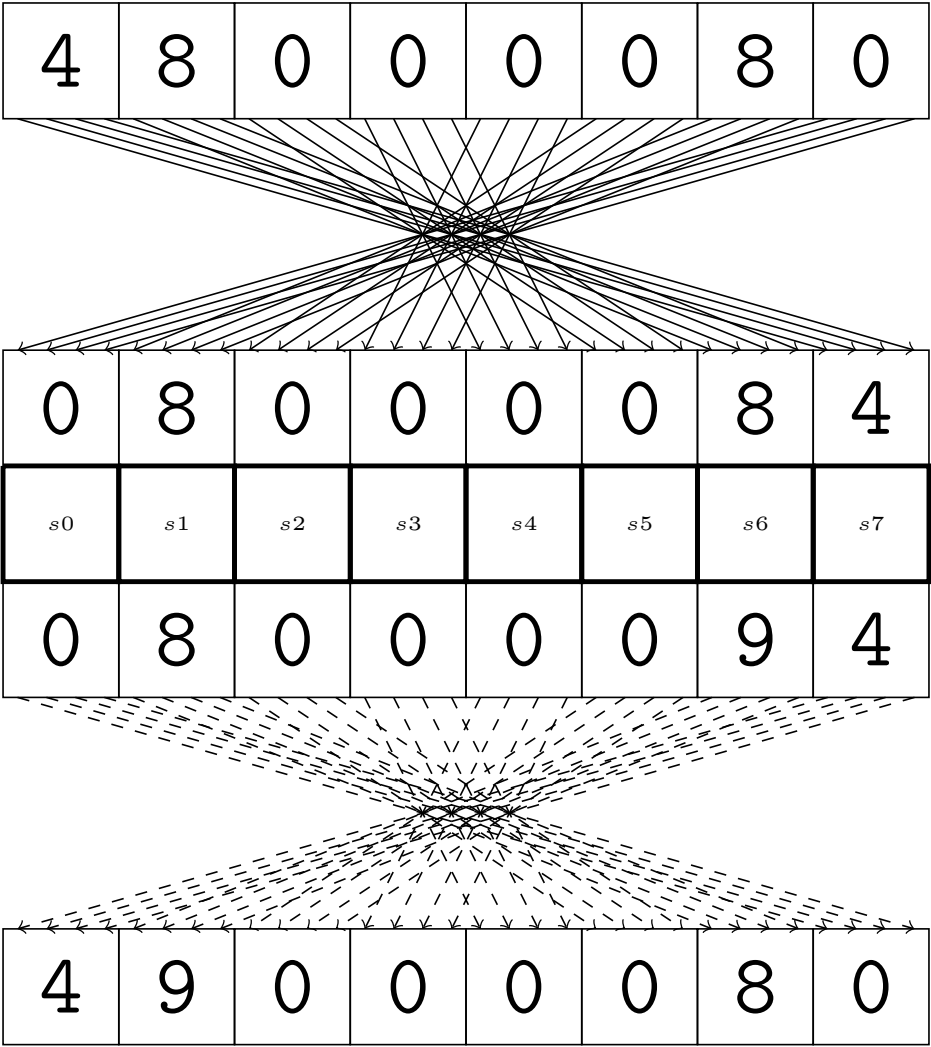
28 LBlock_round

Maximal linear correlation: $2^{-3.0}$

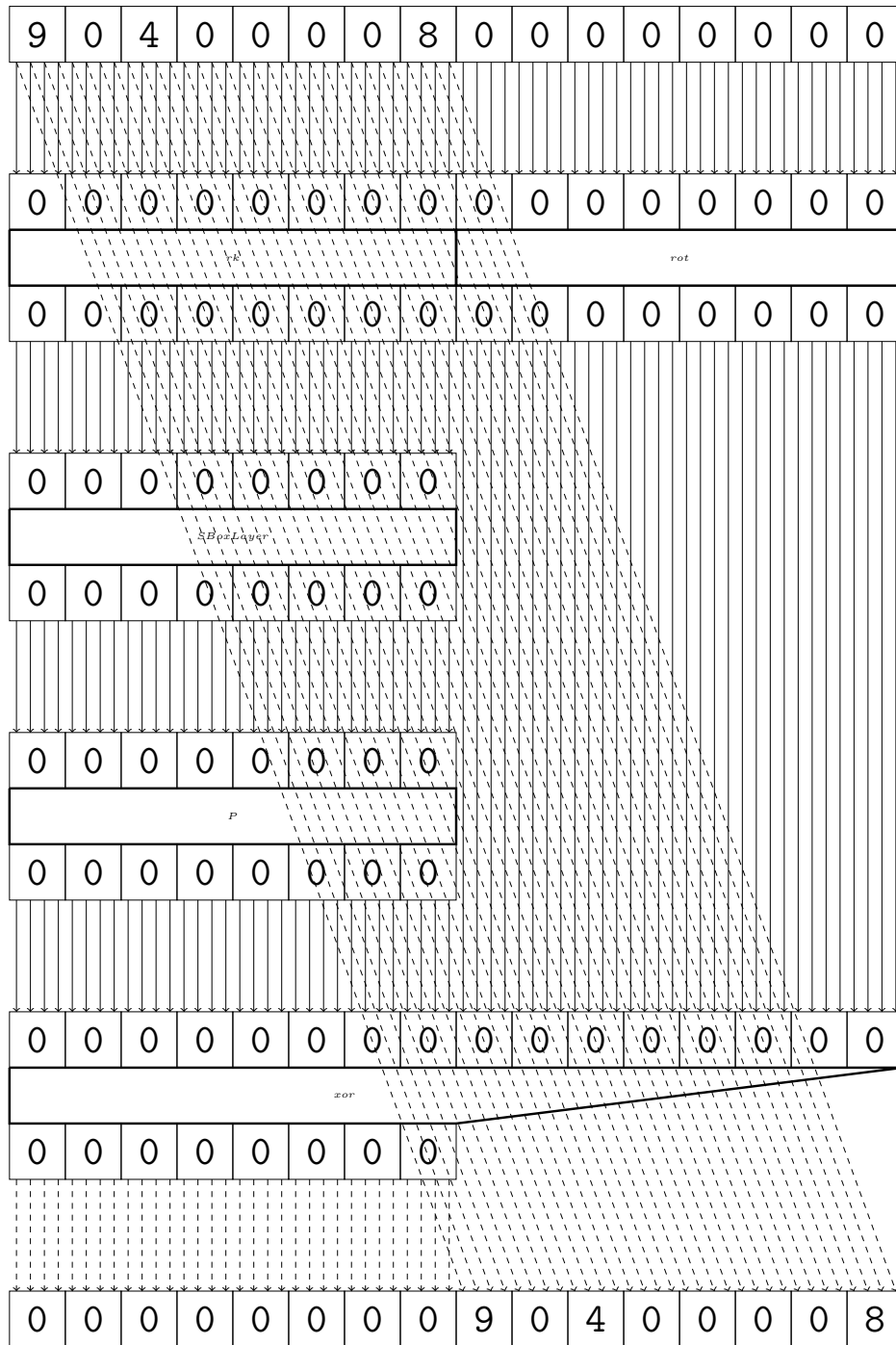


29 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

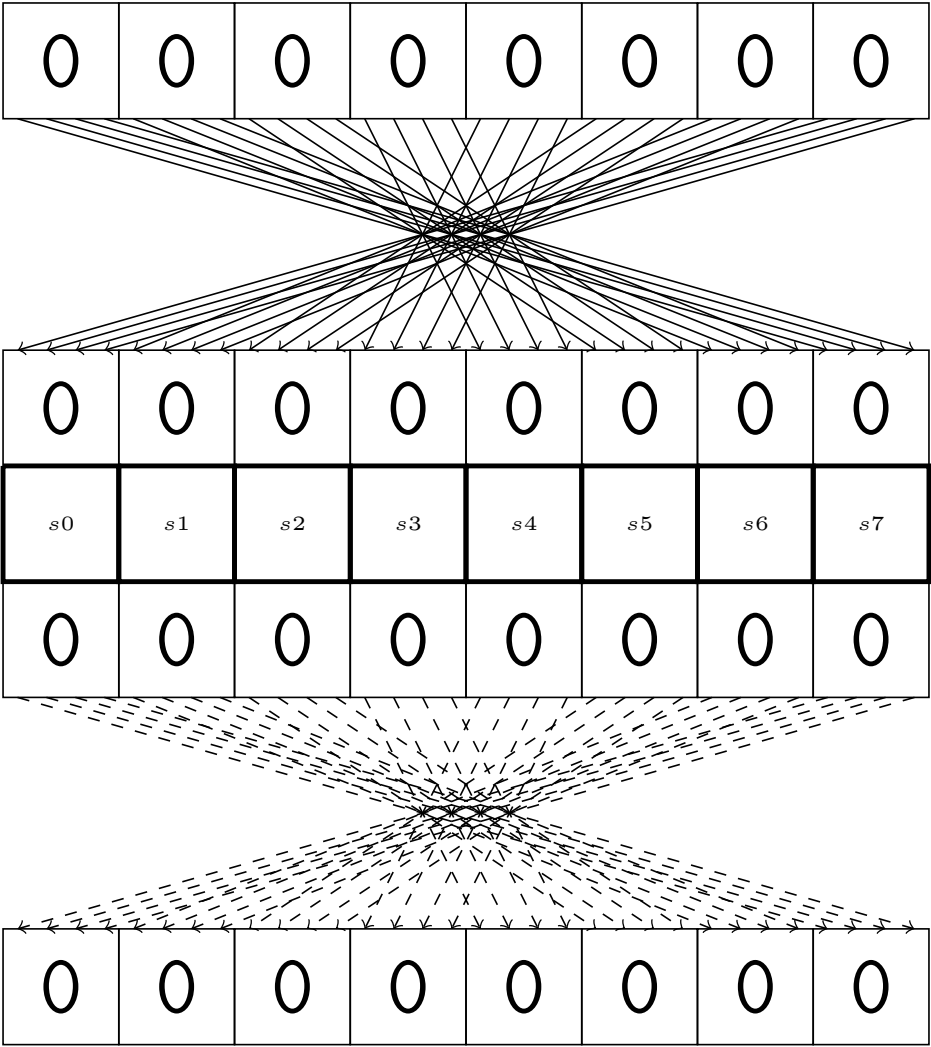


30 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

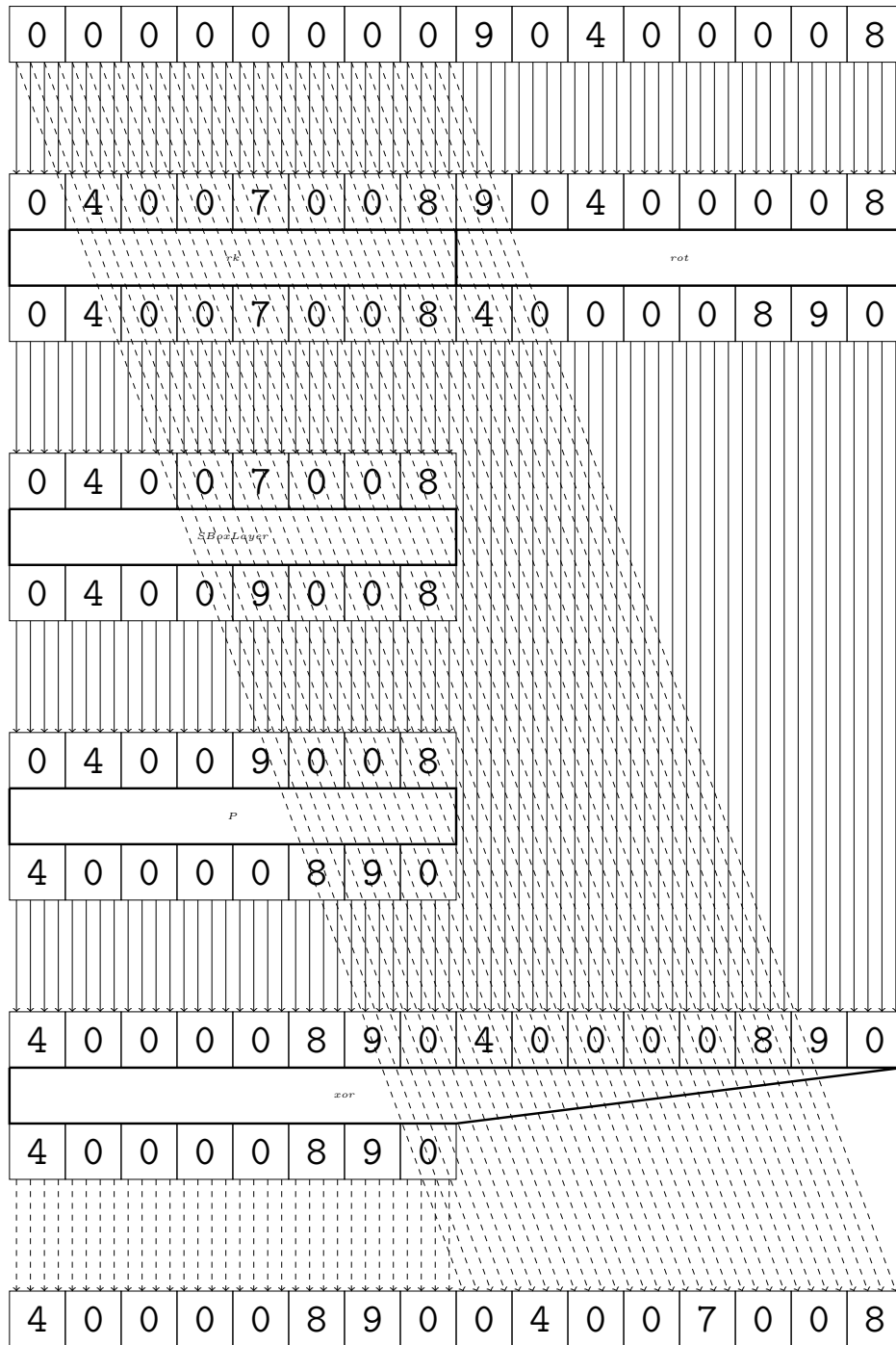
31 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



32 LBlock_round

Maximal linear correlation: $2^{-3.0}$



33 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

